Nicolas Savy-Larigaldie
Gameplay Programmer

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PERSONAL PROJECTS

Kotono Engine (March 2023 - Present)

Technical Programmer | C++, Vulkan, C#, OpenGL

- Created a custom 3D game engine, originally developed in C# using OpenGL, before executing a full architectural rewrite in 2025 to modern C++ and the Vulkan API for hardware-level control and maximum performance.
- Designed a modular multi-project architecture using CMake, separating subsystems, such as rendering, core, reflection, and editor to maximize long term maintainability and drastically reduce build times.
- Engineered subsystems such as a 2D widget system, dynamic 3D entity management, basic physics simulation, and a deferred lighting renderer, peaking at 324,000 updates per second and 1,236 FPS for thousands of objects.
- Implemented a GPU driven rendering pipeline via Vulkan, using dynamic rendering with bindless resource management (descriptor indexing), immensely reducing CPU overhead.

ACADEMIC PROJECTS

Developed multiple prototypes and full-cycle games in multidisciplinary teams (4–10 members) over development periods ranging from 6 weeks to 9 months.

Le Souveniste (October 2025 - June 2026)

Gameplay Programmer & Technical Artist | Unreal Engine 5

- Led gameplay programming and iterative prototyping, developing movement, interaction, and dialogue systems while collaborating directly with the design team to iterate on and refine core mechanics.
- Bridged the gap between gameplay and visuals by implementing custom shader materials, designing animation state machines, and optimizing static and dynamic lighting to match the art team's vision.
- Managed the project's structure and version control, writing convention documents for Git and asset integration, troubleshooting merge conflicts and ensuring smooth asset integration across disciplines.
- Acted as the team's scrum master, organizing weekly stand-up meetings to track individual progress, setting up deadlines for regular builds, and ensuring the project stayed in scope.

DEGREES

Bachelor's Degree in Game Design & Development at e-artsup Toulouse (2023 - 2026)

Official French certification: "Titre RNCP Niveau 6 (n°40867)"

- Specialization: Creative Coding, Gameplay Programming, and Game Architecture.

SKILLS

Languages: C++, C#, GLSL, Python, JavaScript, Dart
Game Engines: Unreal Engine, Unity
Graphics APIs & Tools: Vulkan, OpenGL, Git, Visual Studio, CMake, NVIDIA Nsight
Other Tools: Blender, Adobe Substance Painter/Designer, Photoshop